

1. splice()

Used to **add, remove, or replace** elements in an array.

Syntax:

```
array.splice(start, deleteCount, item1, item2, ...)
```

```
let arr = [10, 20, 30, 40, 50];
```

```
arr.splice(2, 1);
```

```
console.log(arr);
```

```
// [10, 20, 40, 50]
```

- Changes the **original array**.
- 2 → starting index
- 1 → number of elements to delete

Add elements:

```
let arr = [10, 20, 40];
```

```
arr.splice(2, 0, 30);
```

```
console.log(arr);
```

```
// [10, 20, 30, 40]
```

2. slice()

Used to **extract a portion of an array**.

Syntax:

```
array.slice(start, end)
```

```
let arr = [10, 20, 30, 40, 50];
```

```
let result = arr.slice(1, 4);
```

```
console.log(result);
```

```
// [20, 30, 40]
```

- Does **not** change the original array.
- start is included.
- end is excluded.

```
console.log(arr);
```

```
// [10, 20, 30, 40, 50]
```

splice() vs slice()

Method	Original Array	Purpose
splice()	Changes	Add/remove/replace
slice()	Doesn't change	Extract/copy

3. map()

Used to **transform every element** of an array and returns a **new array**.

```
let arr = [1, 2, 3, 4];
```

```
let result = arr.map((x) => x * 2);
```

```
console.log(result);
```

```
// [2, 4, 6, 8]
```

Example:

```
let names = ["prince", "rahul", "jay"];
```

```
let result = names.map((name) => name.toUpperCase());
```

```
console.log(result);
```

```
// ["PRINCE", "RAHUL", "JAY"]
```

Important: map() normally returns an array having the **same number of elements**.

4. reduce()

Used to **combine all array elements into one final value**.

Syntax:

```
array.reduce((accumulator, currentValue) => {
```

```
    return ...;
  }, initialValue);
```

Example: Sum

```
let arr = [10, 20, 30, 40];
```

```
let sum = arr.reduce((total, value) => {
  return total + value;
}, 0);
```

```
console.log(sum);
```

```
// 100
```

How it works:

total = 0

0 + 10 = 10

10 + 20 = 30

30 + 30 = 60

60 + 40 = 100

Another example: Find maximum

```
let arr = [10, 50, 20, 80, 30];
```

```
let max = arr.reduce((a, b) => {
  return a > b ? a : b;
});
```

```
console.log(max);
```

```
// 80
```

Quick Difference

splice() → Add / Remove / Replace

slice() → Extract part of array

map() → Transform every element

reduce() → Convert array into one value

Easy memory trick:

SPLICE = CHANGE

SLICE = COPY

MAP = TRANSFORM

REDUCE = COMBINE